

Model Question Paper-II with effect from 2025-26

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1BMATS201

Second Semester B.E./B.Tech. Degree Examination NUMERICAL METHODS: FOR CSE STREAM

TIME: 03Hours

Max.Marks:100

Note: 1. Answer any FIVE full questions, choosing at least ONE question from each MODULE.

2. VTU Formula Hand Book is Permitted

3. M: Marks, L: Bloom's level, C: Course outcomes.

Module-1			M	L	C
Q 1.	a	Define the following i) Round off Error, ii) Inherent error, iii) Truncation error, iv) Absolute Error, v) Relative error, vi) Percentage error	6	L2	CO-1
	b	Obtain a second degree polynomial approximate to $f(x) = (1+x)^{\frac{1}{2}}$, $x \in [0, 0.1]$ using the Taylor series expansion about $x = 0$. Use the expansion to approximate $f(0.05)$ and find a bound of the truncation error.	7	L3	CO-1
	c	A real root of the equation $f(x) = x^3 - 5x + 1 = 0$ lies in the interval (0,1). Perform four iterations of the Regular-Falsi method to obtain this root.	7	L3	CO-1
OR					
Q 2.	a	Find the absolute error if the number 0.00545828 is i) truncated to 3 decimal digits ii) round off to 3 decimal digits	6	L2	CO-1
	b	Find the root of the equation $x^3 - 4x - 9 = 0$, using the bisection method correct to three decimal places.	7	L3	CO-1
	c	Find the real root of the equation $3x = \cos x + 1$ by Newton Raphson method.	7	L3	CO-1
Module-2					
Q 3.	a	Find the condition number of an invertible matrix $A = \begin{bmatrix} 1 & 3 \\ 3 & 5 \end{bmatrix}$ with respect to L_1 norm.	6	L2	CO-1
	b	Apply LU decomposition to find the solution of the system of linear equations $3x + 2y + 7z = 4$; $2x + 3y + z = 5$; $3x + 4y + z = 7$.	7	L3	CO-1
	c	Using Rayleigh's power method, find the dominant eigen value and corresponding eigen vector for the matrix $\begin{bmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{bmatrix}$ by choosing the initial vector $[1 \ 0 \ 0]^T$. Perform four iterations.	7	L3	CO-1
OR					
Q 4.	a	Show that following system of linear equation is ill conditioned. $2x + y = 4$; $2x + 1.01y = 4$	6	L2	CO-1
	b	Use Gauss Seidel method to approximate the solution of system of linear equations $10x + 2y + z = 9$; $2x + 20y - 2z = -44$; $-2x + 3y + 10z = 22$ near the point (0,0,0). Perform three iterations.	7	L3	CO-1
	c	Using Jacobi's method find an approximate eigen values and eigen vectors of the matrix $\begin{bmatrix} 7 & 1 \\ 1 & 7 \end{bmatrix}$.	7	L3	CO-1

Module-3										
Q 5.	a	From the following table, estimate the number of students who obtained marks between 40 and 45.					6	L3	CO-1	
	Marks :	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80				
	No. of Students:	31	42	51	35	31				
	b	Find the piecewise quadratic interpolation polynomial for the function $f(x)$ defined for the data					7	L3	CO-1	
	x	-2	0	1	3	4				
$f(x)$	-23	1	4	80	193					
Hence interpolate at $x = -0.5$ and $x = 2$.										
c	Find the polynomial $f(x)$ by using Lagrange's formula and hence find $f(3)$ for					7	L3	CO-1		
x :	0	1	2	5						
$f(x)$:	2	3	12	147						
OR										
Q 6.	a	Estimate the value of $f(42)$ using appropriate interpolation formula from the following data.					6	L3	CO-1	
	x :	20	25	30	35	40				45
	$f(x)$:	354	332	291	260	231				204
	b	Evaluate $f(9)$, using Newton's divide difference formula, from the data					7	L3	CO-1	
	x :	5	7	11	13	17				
	$f(x)$:	150	392	1452	2366	5202				
	c	Using piecewise linear interpolation formula to find $f(3)$ and $f(1.5)$ from the data.					7	L3	CO-1	
x :	1	2	4	8						
$f(x)$:	3	7	21	73						
Module-4										
Q 7.	a	Solve $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$					6	L3	CO-2	
	b	Solve $(D^4 + 4D^3 - 5D^2 - 36D - 36)y = 0$					7	L3	CO-2	
	c	Solve $(D^3 + 2D^2 + D)y = e^{-x} + \sin 2x$					7	L3	CO-2	
OR										
Q 8.	a	Solve $(x^3y^2 + xy)dx = dy$					6	L3	CO-2	
	b	Solve $\frac{d^4y}{dx^4} + 8\frac{d^2y}{dx^2} + 16y = 0$					7	L3	CO-2	
	c	Solve $(D^3 + 1)y = \cos(2x - 1) + 3^x$.					7	L3	CO-2	
Module-5										
Q 9.	a	Evaluate $\int_4^{5.2} \log x \, dx$ by 6 equal parts by Simpson's $1/3^{\text{rd}}$ rule					6	L3	CO-2	
	b	Use Modified Euler's method to compute $y(1.2)$ correct to five decimal places given that $\frac{dy}{dx} + \frac{y}{x} = \frac{1}{x^2}$ and $y = 1$ at $x = 1$, taking $h = 0.1$.					7	L3	CO-2	
	c	Using Milne's method, compute y at $x=0.8$ from $\frac{dy}{dx} = x - y^2$ with the data $y(0) = 0, y(0.2) = 0.02, y(0.4) = 0.0795, y(0.6) = 0.1762$.					7	L3	CO-2	
OR										
Q 10.	a	Use Simpson's $3/8$ th rule to find $\int_0^{\pi/2} \sqrt{\cos x} \, dx$ by dividing $[0, \pi/2]$ into six equal parts.					6	L3	CO-2	
	b	Solve by Taylor's series method the equation $\frac{dy}{dx} = \log(xy)$ for $y(1.1)$ and $y(1.2)$ given $y(1)=2$.					7	L3	CO-2	
	c	Use forth order Runge-kutta method to solve $\frac{dy}{dx} = \frac{y-x}{(x+y)}, y(0) = 1$, at $x=0.1$ correct to four decimal places.					7	L3	CO-2	

CBCS SCHEME

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BPHYS102/202

First/Second Semester B.E./B.Tech. Degree Examination, June/July 2026 Quantum Physics and Applications (CSE stream)

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

3. VTU Formula Hand Book is permitted.

Module – 1			M	L	C
Q.1	a.	State Heisenberg's uncertainty principle. Mention its physical significance and illustrate broadening of spectral line using uncertainty principle.	8	L2	CO1
	b.	Set-up Schrodinger's time independent wave equation for a particle in one dimension and mention the same in three dimensions.	8	L2	CO1
	c.	An electron is associated with a de-Broglie wavelength of 1nm. Find the kinetic energy of the electron in eV.	4	L3	CO1
OR					
Q.2	a.	Discuss de-Broglie hypothesis and derive an expression for de-Broglie wavelength. Also discuss the concept of expectation values with its physical significance.	8	L2	CO1
	b.	Find the Eigen functions and energy Eigen values for particle in a one dimensional potential well of infinite height.	8	L2	CO1
	c.	In Planck's constant experiment the knee voltage of a blue LED emitting light wavelength 490nm is 2.54V. Calculate Planck's constant.	4	L3	CO1
Module – 2					
Q.3	a.	Enumerate the assumptions of Quantum free electron theory of metals and discuss the success of Quantum free electron theory of metals.	8	L2	CO2
	b.	Define Fermi level in semiconductor. Show that it always lies mid-way between conduction band and valence band in the case of intrinsic semiconductor.	7	L2	CO2
	c.	Find the temperature at which the probability of occupancy is 0.95 at an energy level 0.3eV below Fermi level.	5	L3	CO2
OR					
Q.4	a.	Describe different electron scattering mechanisms in solids.	8	L2	CO2
	b.	Define Fermi factor and discuss the variation of Fermi factor with temperature and energy.	7	L2	CO2
	c.	The concentration of electrons at room temperature in an extrinsic semiconductor used in Hall experiment is $1.7 \times 10^{23}/\text{m}^3$. If a current of 100 mA flows through the semiconducting strip of thickness 1mm and strength of magnetic field is 3.6 Wb/m^2 , estimate the Hall coefficient and Hall voltage.	5	L3	CO2

Module – 3

Q.5	a.	Explain Silsbee effect. Derive an expression for critical current for a current carrying superconductor wire.	8	L2	CO3
	b.	Explain in detail BCS theory of superconductor and mention the limitations of BCS theory.	7	L2	CO3
	c.	Calculate the critical current for a wire of lead having diameter of 1mm at 4.2K. The critical temperature for lead is 7.18K and critical field is 6.5×10^4 A/m at 0K.	5	L3	CO3

OR

Q.6	a.	What is SQUID? Mention two applications of it. Explain the construction and working of DC SQUID.	8	L2	CO3
	b.	Explain Meissner effect and two fluid model with suitable diagram.	8	L2	CO3
	c.	The critical field of Niobium is 10^4 A/m at 8K and 2×10^5 A/m at 0K. Calculate the transition temperature of the element.	4	L3	CO3

Module – 4

Q.7	a.	Describe the construction and working of semiconductor diode laser.	8	L2	CO4
	b.	What is attenuation in an optical fiber? Discuss the attenuation mechanisms in optical fiber communication.	8	L2	CO4
	c.	In a diffraction grating experiment the distance between screen and grating is 20 cm and the average distance of the second order spot is 2.7 cm. Calculate the wavelength of the laser light. (Grating constant = 1×10^{-5} m.)	4	L3	CO4

OR

Q.8	a.	Explain the construction and working of Mach-Zehnder interferometer. Mention its applications.	8	L2	CO4
	b.	What are optical modulators? Explain Pockels effect and Kerr effect.	7	L2	CO4
	c.	The sum of refractive indices of core and cladding in a step index optical fiber is 3.12 and their difference is 0.08. The diameter of the core is $50 \mu\text{m}$. Find V- number at wavelength 594 nm.	5	L3	CO4

Module – 5

Q.9	a.	Mention any two properties of a qubit. With a neat diagram explain the representation of qubits using Bloch sphere.	8	L2	CO5
	b.	Discuss the working of T gate with its matrix representation and action on $ 0\rangle$ and $ 1\rangle$ states.	8	L2	CO5
	c.	Show that Hadamard gate is unitary.	4	L3	CO5

OR

Q.10	a.	Explain anharmonicity along with a relevant diagram and its importance in quantum computing.	8	L2	CO5
	b.	Explain the operation of CNOT gate on four two qubits. Mention truth table and matrix representation.	8	L2	CO5
	c.	A linear operator 'X' operates such that $X 0\rangle = 1\rangle$ and $X 1\rangle = 0\rangle$. Find matrix representation of 'X'.	4	L3	CO5

Second Semester B.E./B.Tech. Degree Examination, June/July 2026

Numerical Methods : CSE Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. M : Marks, L: Bloom's level, C: Course outcomes.
 3. VTU Formula Hand Book is permitted.*

Module - 1			M	L	C
Q.1	a.	Round off the number 865250 to four significant figures and compute absolute error, relative error and percentage error.	6	L2	CO1
	b.	Find a real root of the equation $xe^x - 2 = 0$ correct to three decimal places using Newton-Raphson method. (carryout 3 iterations)	7	L3	CO1
	c.	Find a root of $x^3 - 4x + 1 = 0$ using Bisection method. (carryout 5 iterations).	7	L3	CO1
OR					
Q.2	a.	Find the absolute error if the number $X = 0.00545828$ is i) Truncated to 3 decimal digits ii) Rounded off to 3 decimal digits	6	L2	CO1
	b.	Find a real root of the equation $\cos x - 3x + 1 = 0$ by using Regula-Falsi method.	7	L3	CO1
	c.	Using Secant method, find a real root of the equation $x^3 - 2x + 5 = 0$, correct to three decimal places. (carryout 4 iterations).	7	L3	CO1
Module - 2					
Q.3	a.	If $V = (0, -1, 1, 2)$ and $W = (1, 5, 1, 8)$ compute L_1, L_2, L_∞ norms of V, W and $2V+W$.	6	L2	CO1
	b.	Find the dominant Eigen value and the corresponding Eigen vector of the matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ by power method taking the initial Eigen vector as $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.	7	L3	CO1
	c.	Using Jacobi's Method to find the Eigen values and the corresponding Eigen vectors of the matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$	7	L3	CO1
1 of 3					

OR

Q.4	a.	Show that the system of equations ill conditioned $x + 3y = 4$, $2x + 5y + 7$.	6	L2	CO1
	b.	Apply Gauss-Seidel iteration method upto 3 iteration to solve $10x + y + z = 12$, $x + 10y + z = 12$, $x + y + 10z = 12$	7	L3	CO1
	c.	Use LU decomposition method to solve the following system of equations $2x + y + 4z = 12$, $4x + 11y - z = 33$, $8x - 3y + 2z = 20$.	7	L3	CO1

Module - 3

Q.5	a.	The area of a circle (A) corresponding to diameter (D) is given below <table><tr><td>D</td><td>80</td><td>85</td><td>90</td><td>95</td><td>100</td></tr><tr><td>A</td><td>5026</td><td>5674</td><td>6362</td><td>7088</td><td>7854</td></tr></table> Find the area corresponding to diameter 105 using an interpolation formula.	D	80	85	90	95	100	A	5026	5674	6362	7088	7854	7	L3	CO1
D	80	85	90	95	100												
A	5026	5674	6362	7088	7854												
	b.	Use Lagrange's interpolation formula to find $f(4)$ given <table><tr><td>x</td><td>0</td><td>2</td><td>3</td><td>6</td></tr><tr><td>f(x)</td><td>-4</td><td>2</td><td>14</td><td>158</td></tr></table>	x	0	2	3	6	f(x)	-4	2	14	158	7	L3	CO1		
x	0	2	3	6													
f(x)	-4	2	14	158													
	c.	Obtain the Piecewise linear interpolating polynomial defined by the data. <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>f(x)</td><td>1</td><td>2</td><td>5</td><td>10</td></tr></table> And hence find the value of y at $x = 0.5$ & 1.5.	x	0	1	2	3	f(x)	1	2	5	10	6	L3	CO1		
x	0	1	2	3													
f(x)	1	2	5	10													

OR

Q.6	a.	From the following table find the number of students who obtained marks between 40 and 45 <table border="1"> <tr> <td>Marks</td><td>30-40</td><td>40-50</td><td>50-60</td><td>60-70</td><td>70-80</td></tr> <tr> <td>No. of students</td><td>31</td><td>42</td><td>51</td><td>35</td><td>31</td></tr> </table>	Marks	30-40	40-50	50-60	60-70	70-80	No. of students	31	42	51	35	31	7	L3	CO1
Marks	30-40	40-50	50-60	60-70	70-80												
No. of students	31	42	51	35	31												
	b.	Given $u_0 = -5$, $u_1 = -14$, $u_4 = -125$, $u_8 = -21$, $u_{10} = 355$ by using Newton's divided difference formula to find u_2 .	7	L3	CO1												
	c.	Obtain Piecewise interpolating polynomial and hence find $f(2.0)$ <table border="1"> <tr> <td>x</td><td>-2</td><td>0</td><td>1</td><td>3</td><td>4</td></tr> <tr> <td>y</td><td>-23</td><td>1</td><td>4</td><td>82</td><td>193</td></tr> </table>	x	-2	0	1	3	4	y	-23	1	4	82	193	6	L3	CO1
x	-2	0	1	3	4												
y	-23	1	4	82	193												

Module - 4

Q.7	a.	Solve $(y^3 - 3x^2y)dx - (x^3 - 3xy^2)dy = 0$	6	L3	CO2
	b.	Solve $(4D^4 - 4D^3 - 23D^2 + 12D + 36)y = 0$	7	L3	CO2
	c.	Solve $(D^3 - D^2 + 4D - 4)y = \sinh(2x + 3)$	7	L3	CO2

OR

Q.8	a.	Solve $\frac{dy}{dx} - \frac{1}{2}\left(1 + \frac{1}{x}\right)y + \frac{3y^3}{x} = 0$	6	L3	CO2
	b.	Solve $(x^2 + y^2 + x)dx + xydy = 0$	7	L3	CO2
	c.	Solve $(D^3 - 1)y = 3\cos 2x$	7	L3	CO2

Module - 5

Q.9	a.	Use Simpson's $\frac{1}{3}$ rd rule with six equal parts to evaluate $\int_0^6 3x^2 dx$.	6	L3	CO1
	b.	Using Runge-Kutta method of 4 th order to find $y(0.2)$ for the equation $\frac{dy}{dx} = \frac{y-x}{y+x}$, $y(0) = 1$ taking $h = 0.2$.	7	L3	CO1
	c.	Apply Milne's method to compute $y(1.4)$ correct to 4 decimal places given $\frac{dy}{dx} = x^2 + \frac{y}{2}$ and the following data $y(1) = 2$, $y(1.1) = 2.2156$, $y(1.2) = 2.4649$, $y(1.3) = 2.7514$.	7	L3	CO1

OR

Q.10	a.	By using Weddle's rule with 6 equal strips, find the value of $\int_0^1 \frac{x}{1+x^2} dx$	6	L3	CO1
	b.	Apply Taylor's series method to find y at $x = 0.1$ given $\frac{dy}{dx} = 2y + 3e^x$ & $y(0) = 0$.	7	L3	CO1
	c.	Using modified Euler's method find $y(20.2)$ & $y(20.4)$ given that $\frac{dy}{dx} = \log_{10}\left(\frac{x}{y}\right)$ with $y(20) = 5$ taking $h = 0.2$.	7	L3	CO1
